Badge Hacking for Beginners

Nathan Jones

November 2026



https://lightturtles.github.io

LightTurtles

Welcome!

- Make sure you have a way to connect to the badge. Either:
  - Install Thonny
  - Use the browser-based Viper IDE with a Chrome-based web browser (Chrome, Edge, Brave, Opera)
  - Install `mpremote` (if you prefer the command line)
- Clone the badge's firmware repo
- Get an I2C sensor and a cable (graciously donated by DigiKey !)
- Get to know the folks around you!



https://lightturtles.github.io

The Plan

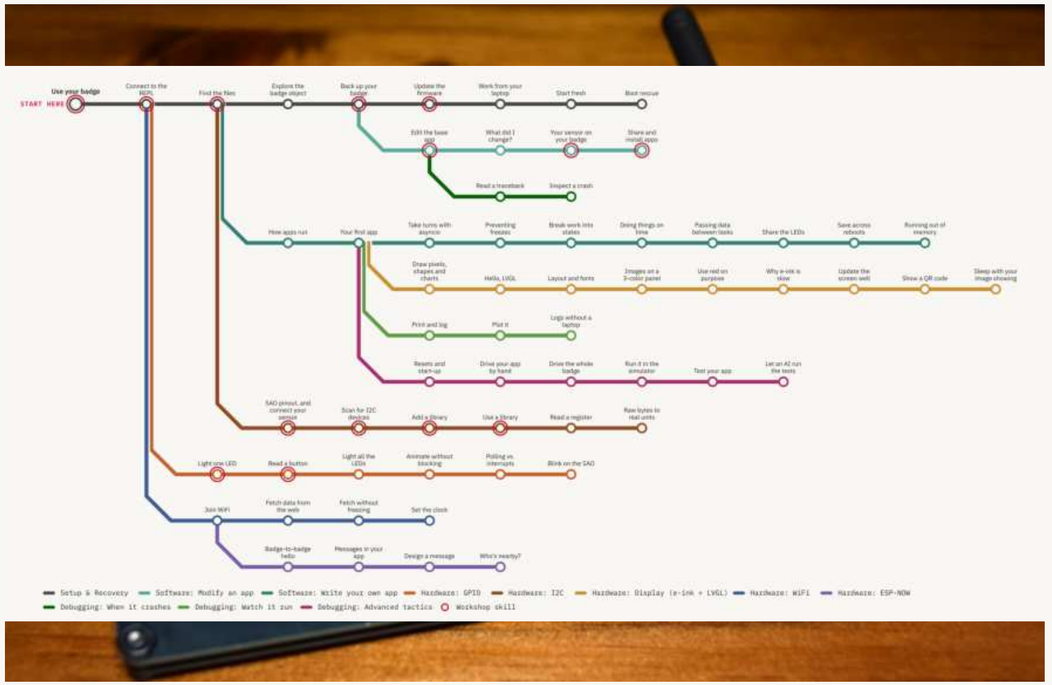
1. Intro (20 min)
2. Skills map (90 min)
    1. 14 skills
    2. The result
3. Wrap-up (10 min)
    1. Plan your hack!
    2. Provide some feedback



https://lightturtles.github.io

The Plan

1. Intro (20 min)
2. Skills map (90 min)
    1. 14 skills
    2. The result
3. Wrap-up (10 min)
    1. Plan your hack!
    2. Provide some feedback



https://lightturtles.github.io

What was the first electronic project you built yourself?

https://lightturtles.github.io

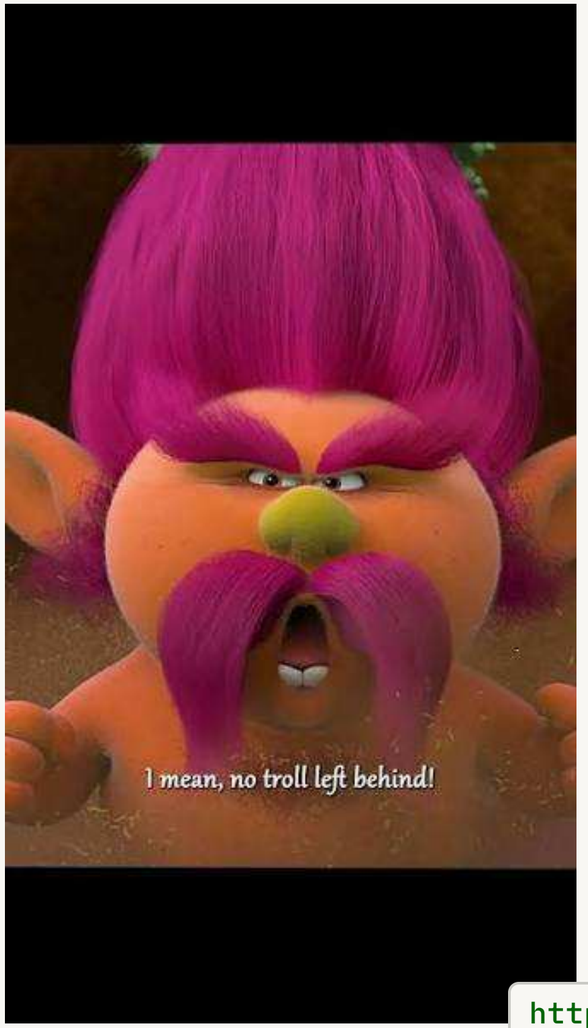
Nathan Jones

https://lightturtles.github.io



- Master's of Computer Engineering from NC STATE UNIVERSITY
- 15 years in the U.S.ARMY as a Military Intelligence Officer
  - 5 of those at West Point
- Selected works
  - "Enter the Matrix"
  - "Inside the Voja4"
  - "How Hardware Gets Hacked"
- Owner/operator LightTurtles, LLC Embedded Systems Design & Education (lightturtles.github.io)

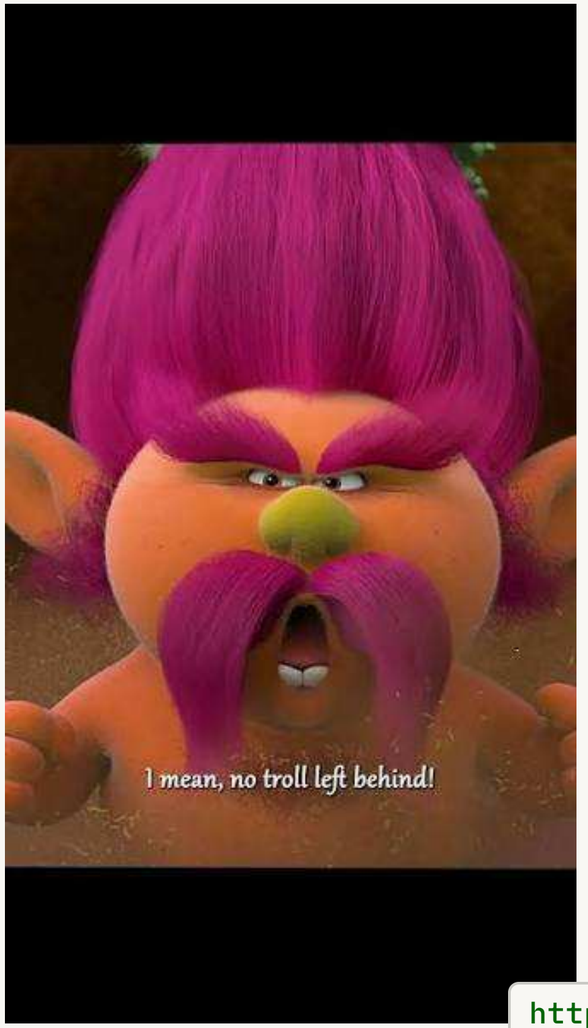
Three rules



https://lightturtles.github.io

Three rules

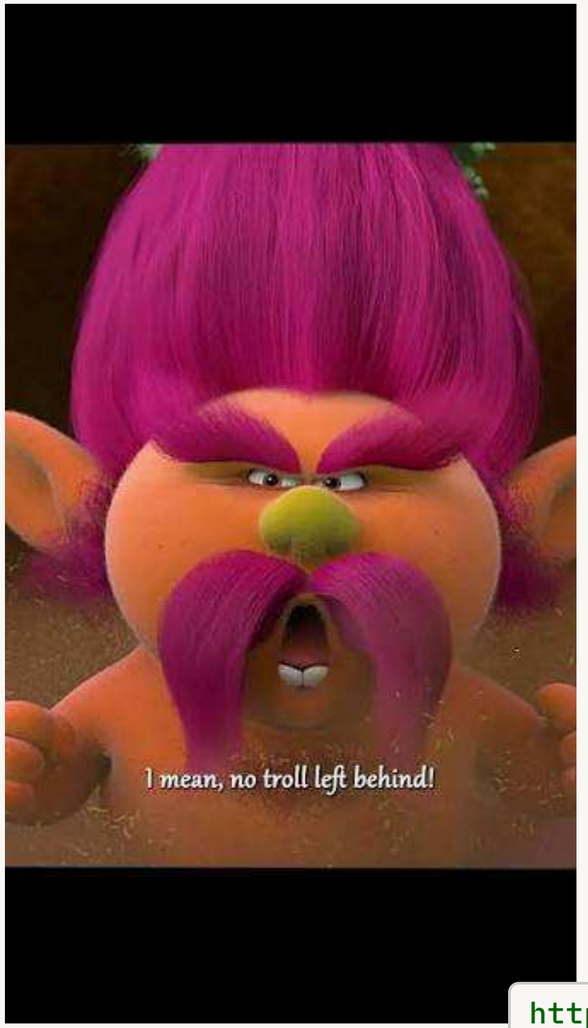
- No troll left behind!



https://lightturtles.github.io

Three rules

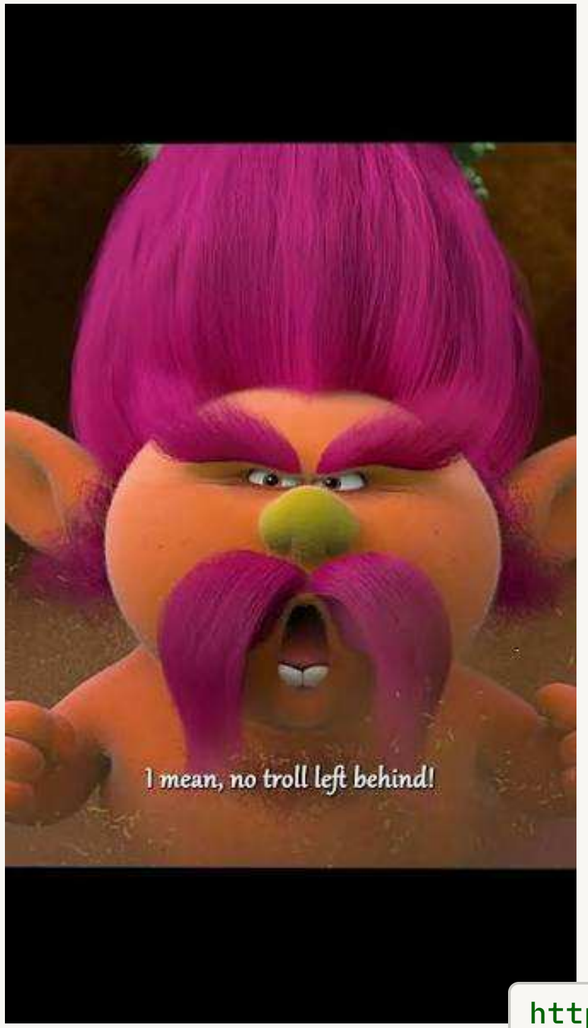
- **No troll left behind!**
  - Check out what I just discovered!



https://lightturtles.github.io

Three rules

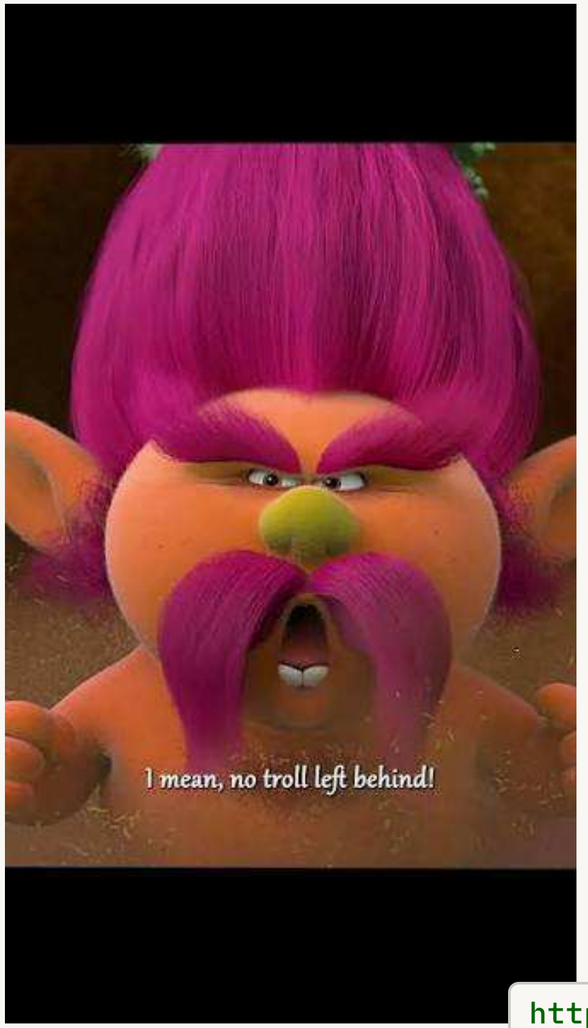
- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?



https://lightturtles.github.io

Three rules

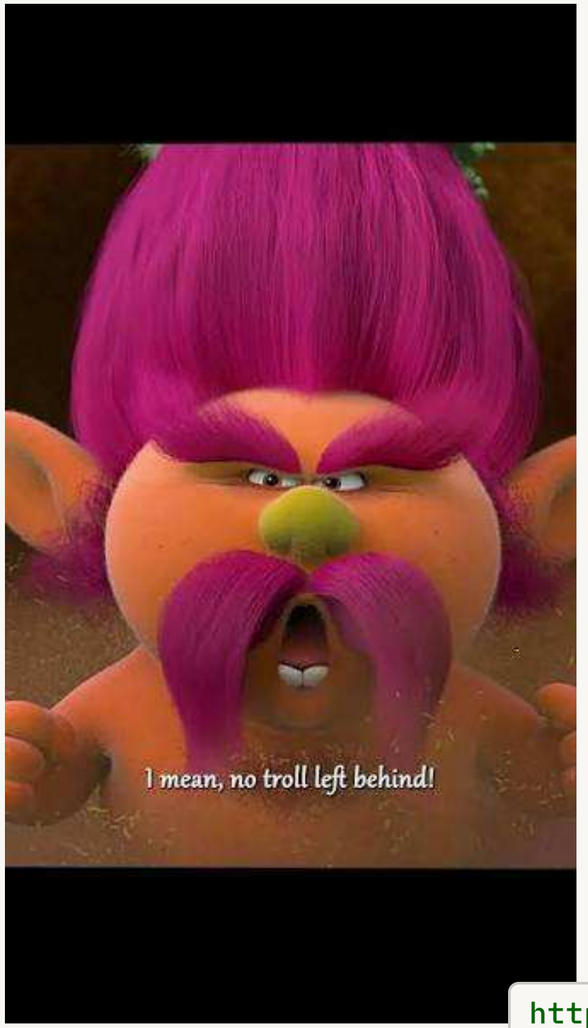
- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?
- **Be curious/creative/subversive**



https://lightturtles.github.io

Three rules

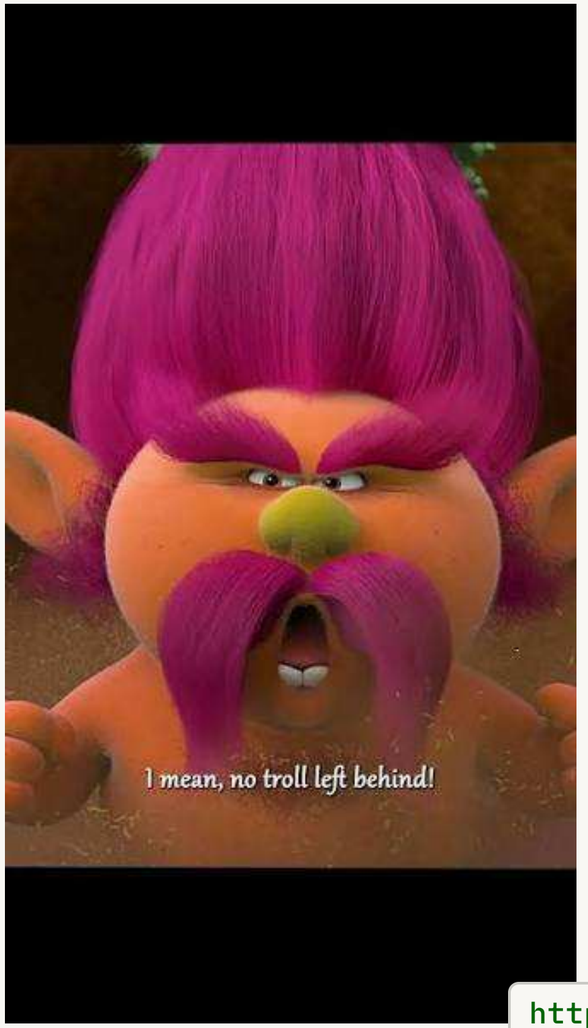
- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?
- **Be curious/creative/subversive**
  - I wonder what would happen if...



https://lightturtles.github.io

Three rules

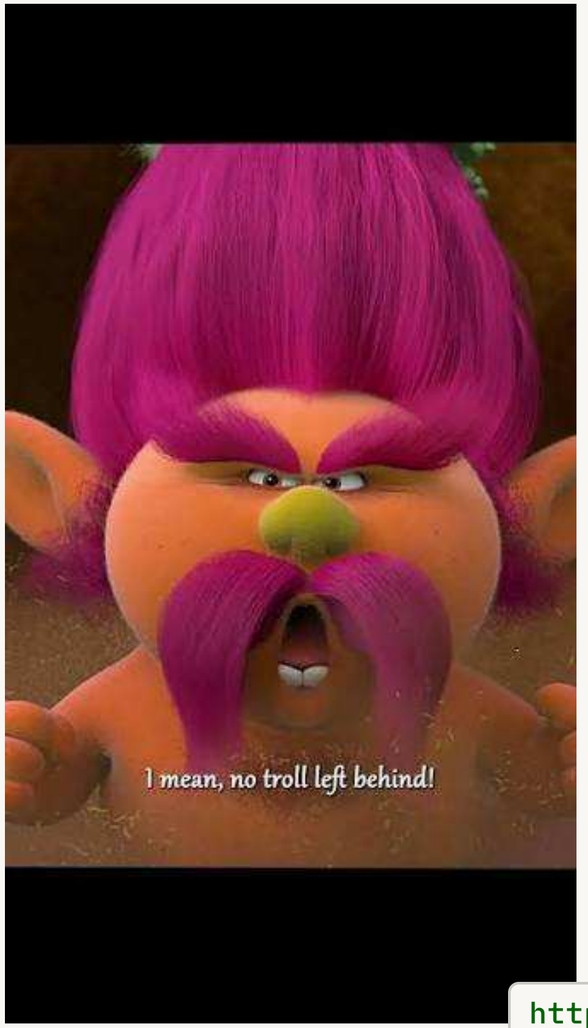
- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?
- **Be curious/creative/subversive**
  - I wonder what would happen if...
  - Wait, what if I...



https://lightturtles.github.io

Three rules

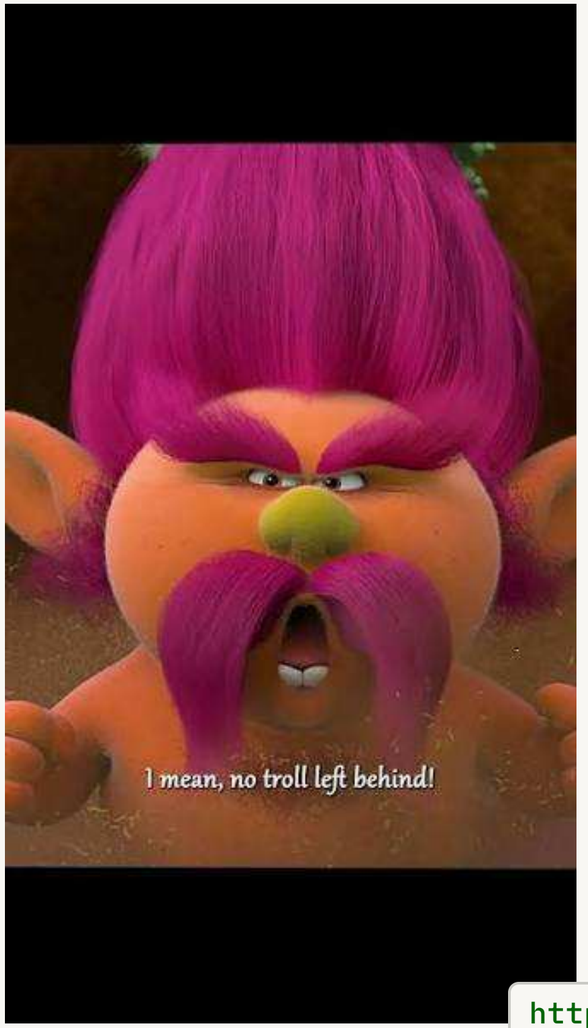
- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?
- **Be curious/creative/subversive**
  - I wonder what would happen if...
  - Wait, what if I...
- **Periodically stop and summarize**



https://lightturtles.github.io

Three rules

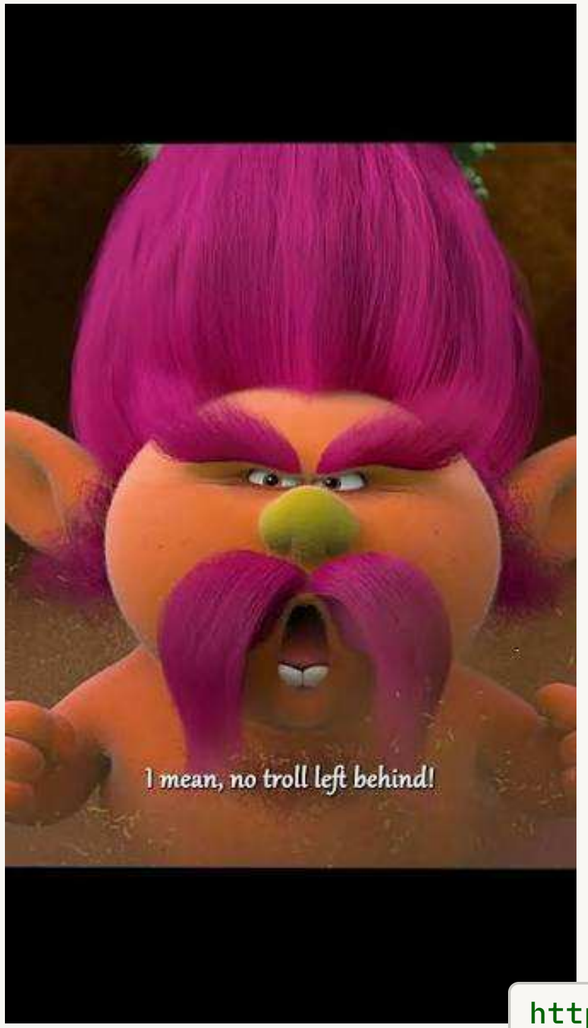
- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?
- **Be curious/creative/subversive**
  - I wonder what would happen if…
  - Wait, what if I…
- **Periodically stop and summarize**
  - So doing X makes Y happen.



https://lightturtles.github.io

Three rules

- **No troll left behind!**
  - Check out what I just discovered!
  - Wait, what are you doing there?
- **Be curious/creative/subversive**
  - I wonder what would happen if…
  - Wait, what if I…
- **Periodically stop and summarize**
  - So doing X makes Y happen.
  - Here's a graphic I drew about how this works.



https://lightturtles.github.io

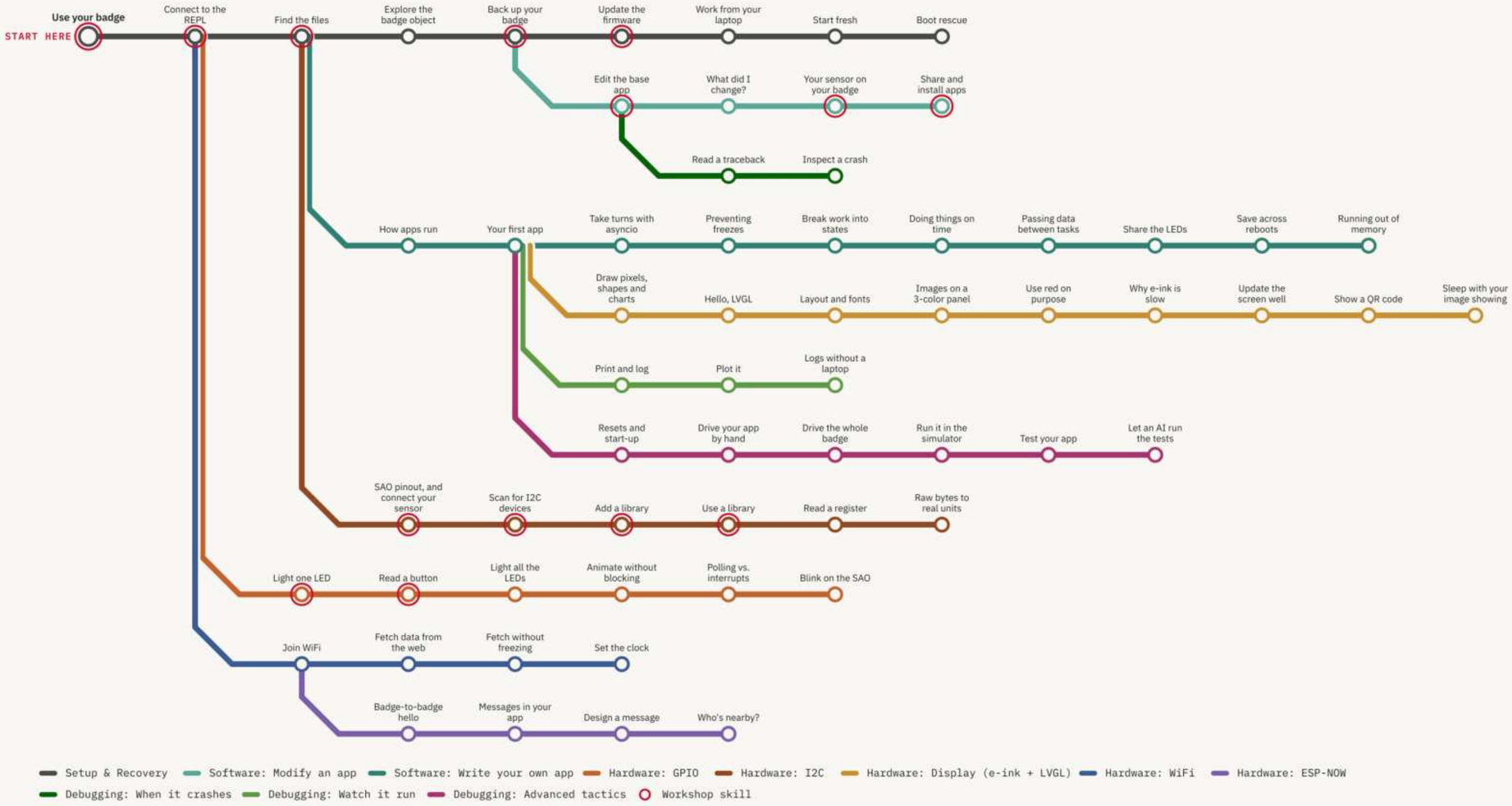
The 2026 Supercon Badge



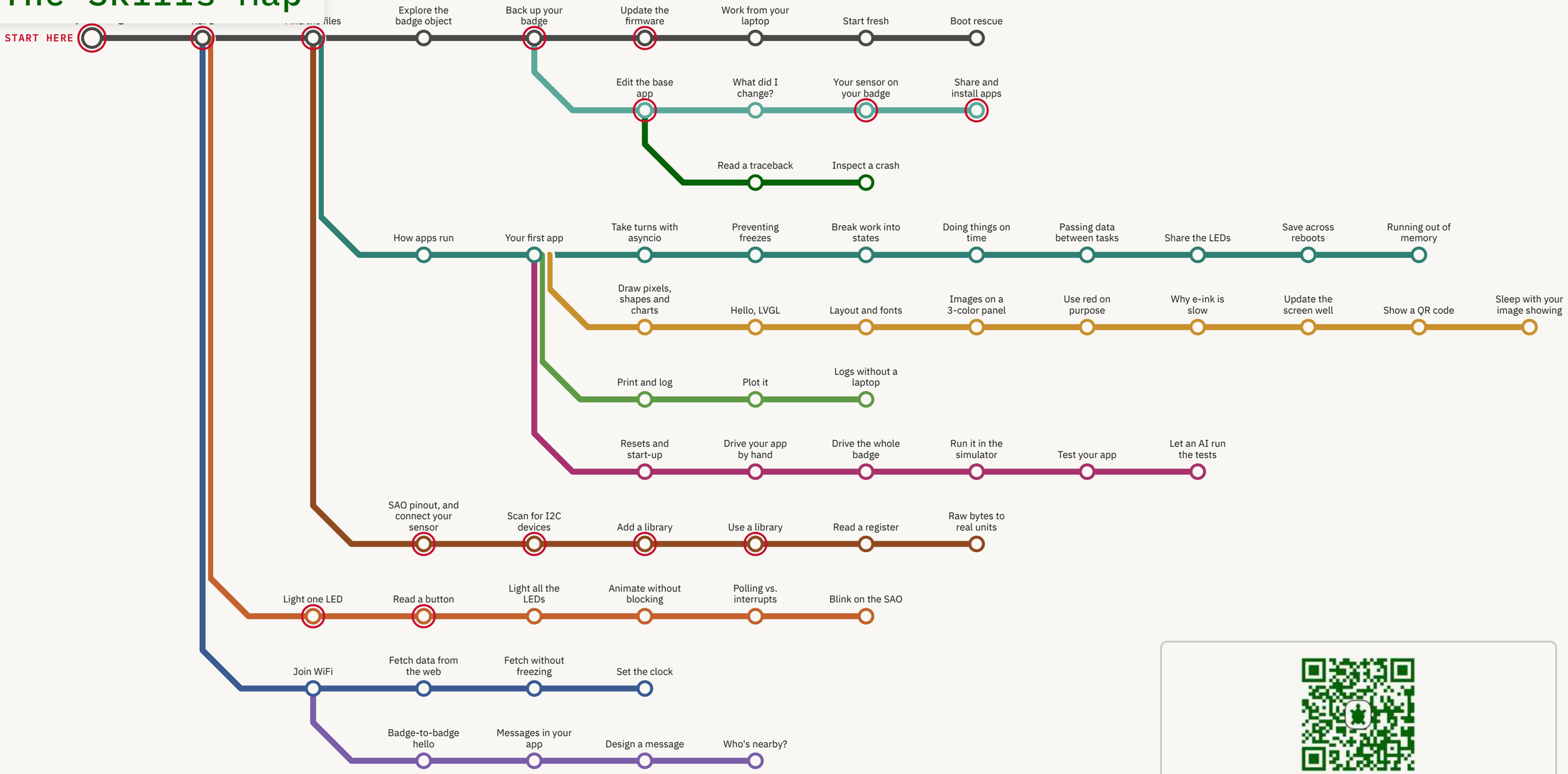
https://lightturtles.github.io

The Skills Map

https://lightturtles.github.io/play/supercon26

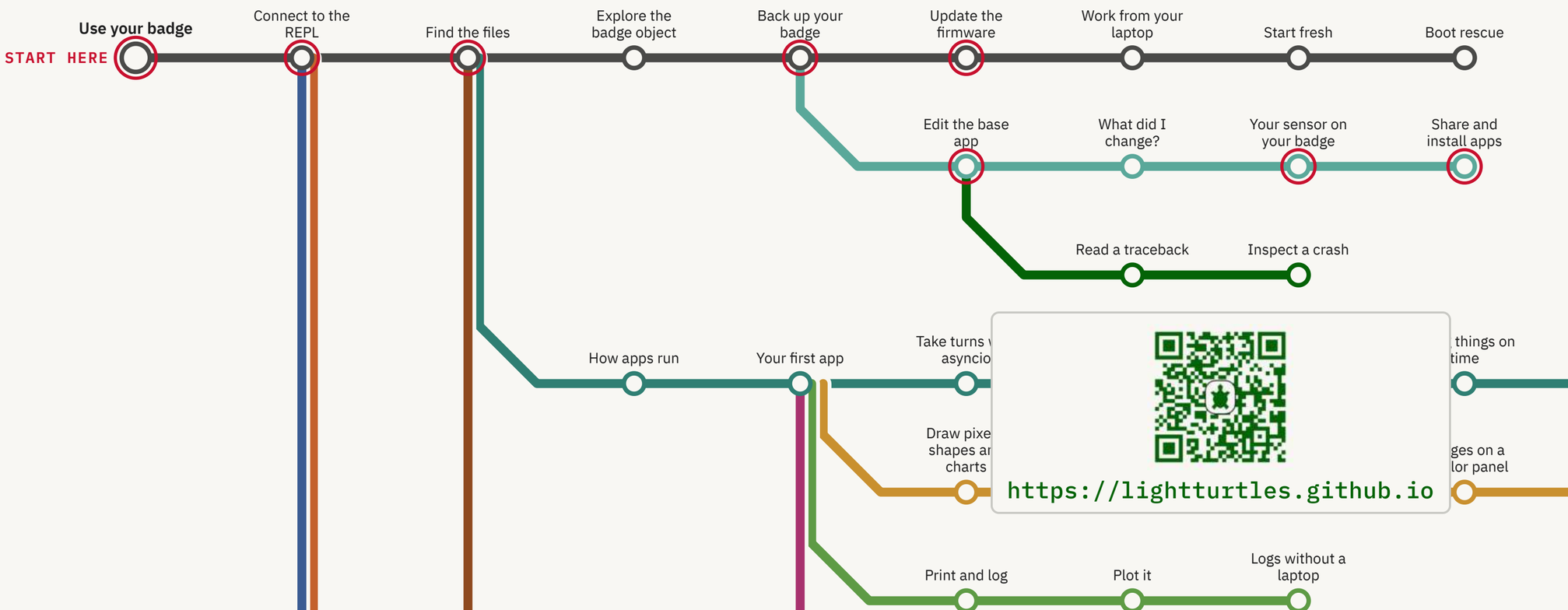


The Skills Map



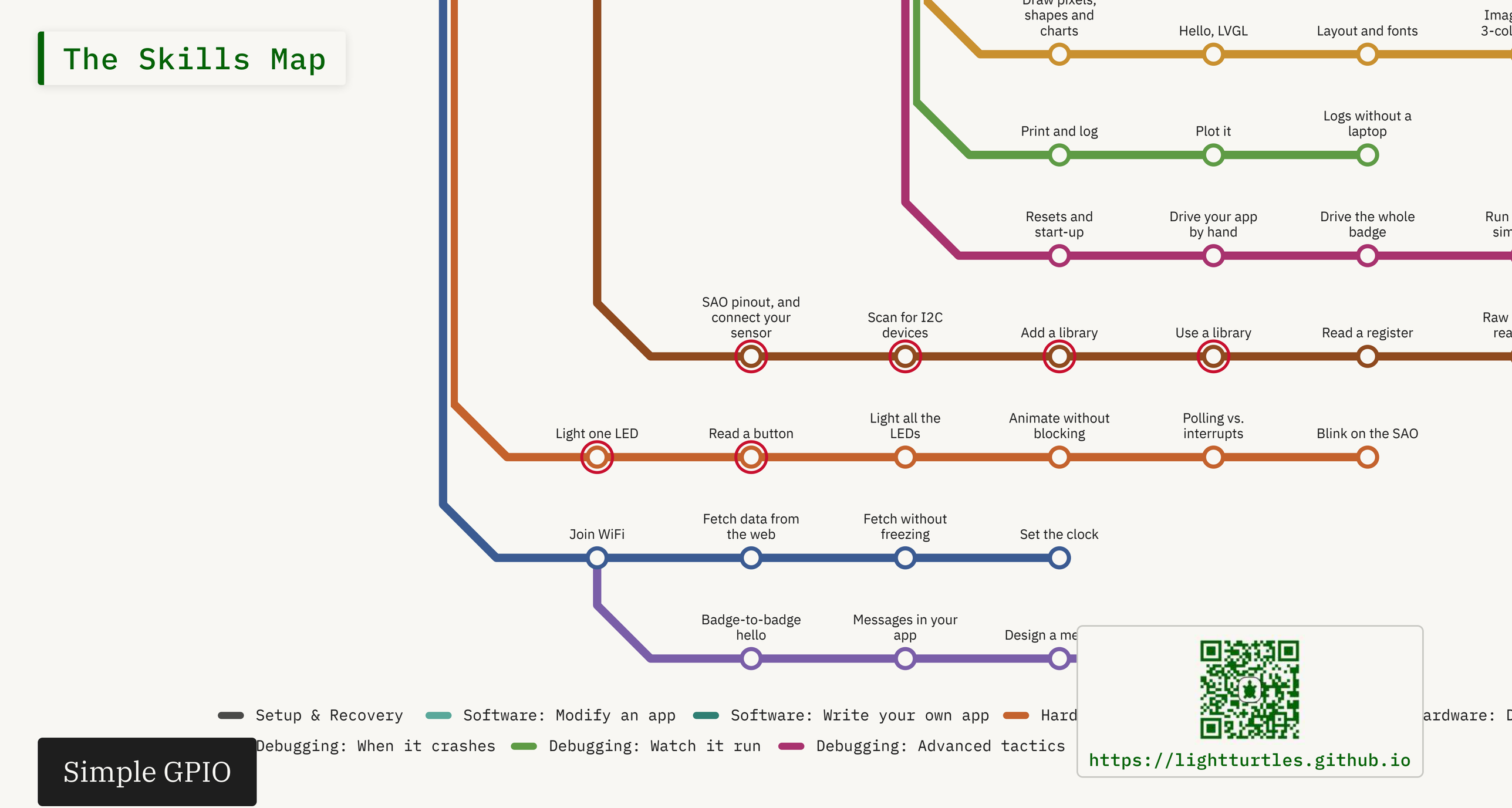
https://lightturtles.github.io

The Skills Map



Badge orientation

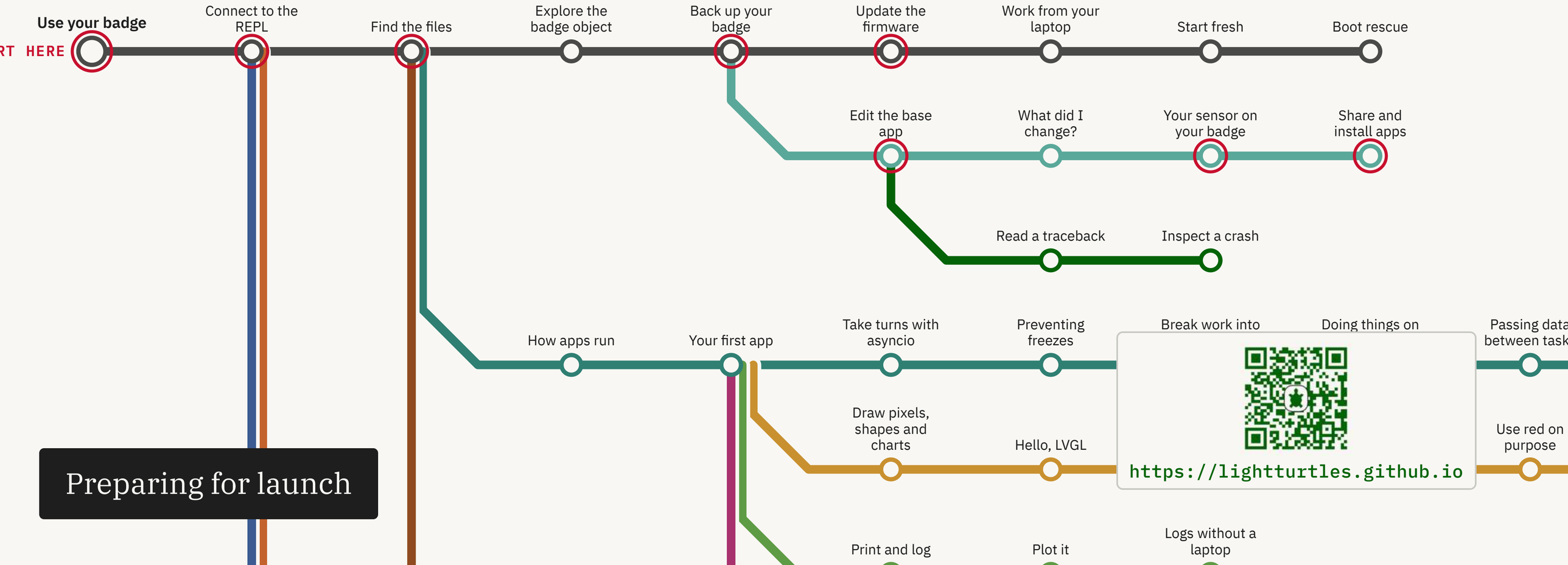
The Skills Map



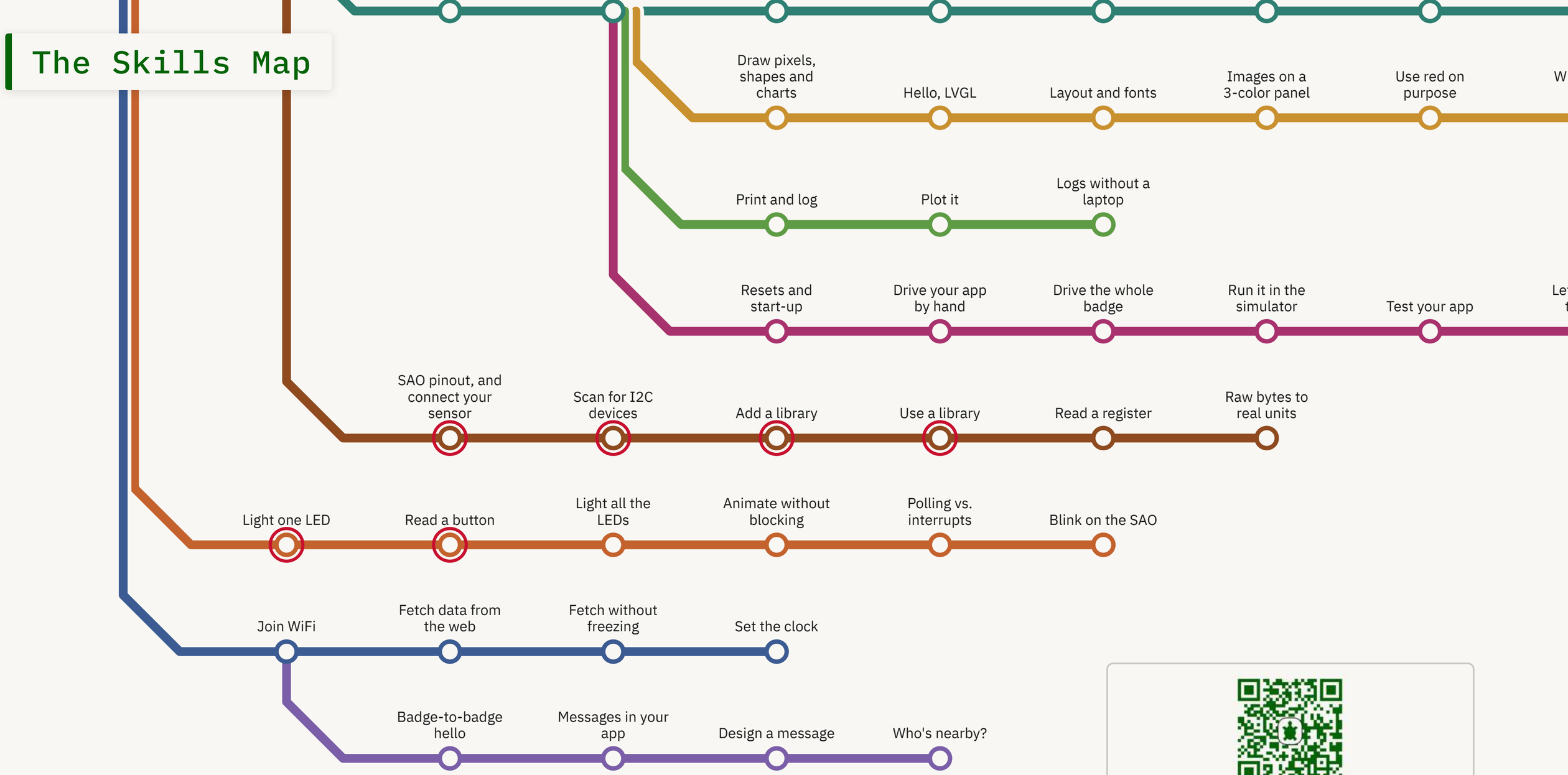
https://lightturtles.github.io

Simple GPIO

The Skills Map



The Skills Map



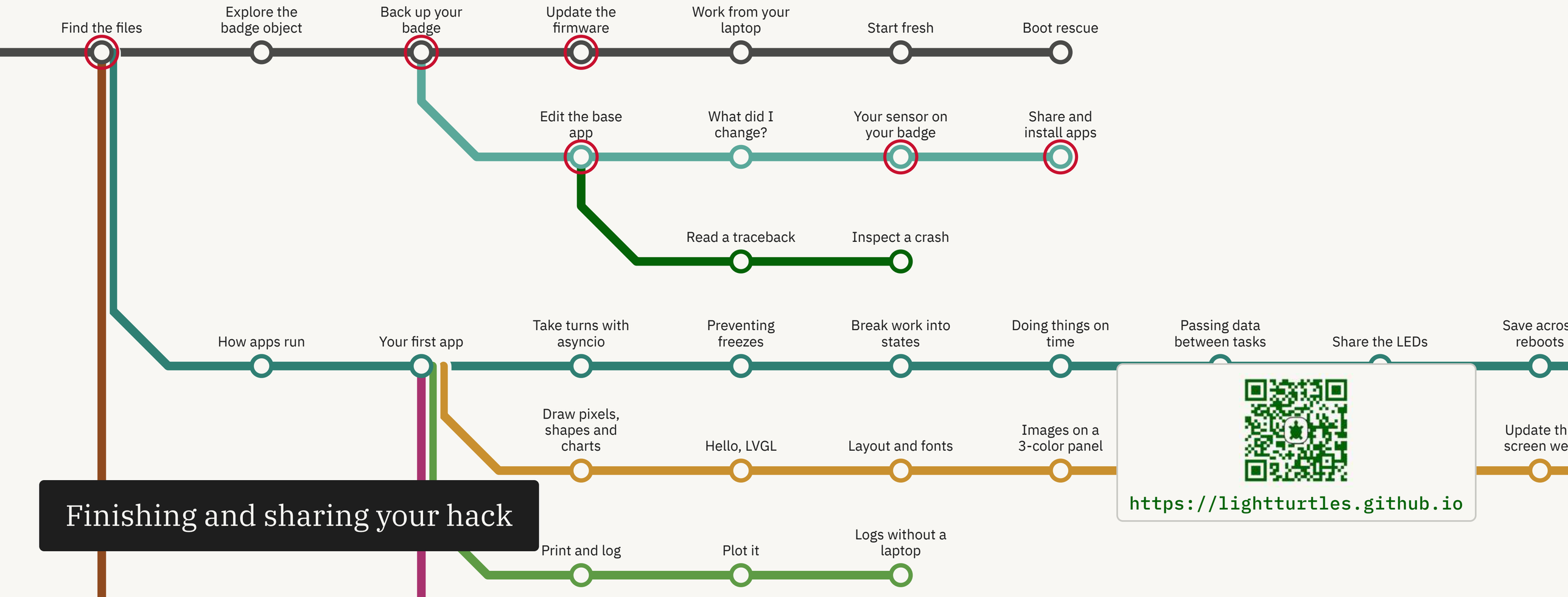
I2C sensors

https://lightturtles.github.io

: Modify an app — Software: Write your own app — Hardware: GPIO — Hardware: I2C — Hardw

Debugging: Watch it run — Debugging: Advanced tactics — Workshop skill

The Skills Map



Diagrams?

https://lightturtles.github.io

Survey

https://lightturtles.github.io